

TINGXI LI

800 W Campbell Road, Richardson, TX 75080

☎ (213)-795-5275 ✉ tingxi.li@utdallas.edu [in linkedin](#) [🌐 homepage](#) [🎓 Google Scholar](#)

Education

University of Texas at Dallas

Doctor of Philosophy in Computer Science

Sep. 2024 – Present

Dallas, TX

Dalian University of Technology

Bachelor of Science

Sep. 2019 – May 2024

Dalian, China

Academic Experience

Adversarial Attack on a Robotic Arm

UT Dallas

Feb. 2025 – Present

Dallas, TX

- Analyzed robustness in Bulletarm grasping tasks, revealing deterministic environment dynamics and non-robust action-space boundaries that can be exploited by adversarial environment settings.

Efficiency Attack on Multi-level Applications

UT Dallas

May 2025 – Present

Dallas, TX

- Applied efficiency attacks on multi-level applications, evaluating their impact on downstream tasks by measuring energy consumption and time delays.
- Proposed and developed defense mechanisms to detect and mitigate these attacks, enhancing system resilience.

GPU Kernel Code Generation

UT Dallas

Oct. 2025 – Present

Dallas, TX

- Researching machine learning compilation techniques to automate GPU kernel code generation using large language models, aiming for semantic correctness and near state-of-the-art efficiency.
- Investigating sources of non-determinism in GPU kernel execution and developing compiler-level solutions for deterministic and reproducible performance.

Internship Experience

Sophgo

Research Intern

June. 2024 – Aug. 2024

Shenzhen, China

- Developed and implemented API code in C++ for models deployed on RISC-V processors. Authored comprehensive documentation to support efficient deployment and usage.
- Fine-tuned models on private datasets, identified failure cases, and implemented data augmentation strategies.

Projects

Amazon Trusted AI Competition

Team Member

Nov. 2024 – Jul. 2025

Seattle, WA

- Finalist team in the Amazon Trusted AI Competition, focused on jailbreaking CodeLLMs under a black-box setting, awarded \$250K prize.
- Built an iterative model refinement pipeline to continuously integrate new attack data and improve surrogate model accuracy. Fine-tuned surrogate models to evaluate and guide adversarial prompt generation, achieving 0.80+ recall and 0.75+ F1-score in malicious intent detection.

Publications

SoK: Efficiency Robustness of Dynamic Deep Learning Systems

USENIX Security 2025

Ravishka Rathnasuriya, **Tingxi Li**, Zexin Xu, Zihe Song, Mirazul Haque, Simin Chen, Wei Yang

- Co-authored a systematization-of-knowledge (SoK) paper surveying efficiency attacks on dynamic DNN systems. Categorized attack strategies, reproduced published results, and benchmarked defenses in unstudied scenarios.

COMET: Closed-loop Orchestration for Malicious Elicitation Techniques in Code Models

Zexin Xu, **Tingxi Li**, et al.

- Co-authored a public technical report detailing the attack and defense methodology, outlining surrogate model fine-tuning and iterative adversarial data integration pipeline.

Miscellaneous

Research Interest: AI Security; Software Engineering; Deep Learning Compilers

Tech Stack: Python; C; C++; Java; PyTorch; LaTeX; SQL